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Palaeomagnetism of the permo/carboniferous (280 Ma) Jebel Nehoud ring complex, Kordofan, Central Sudan

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Abstract

Palaeogeographic reconstructions of Pangaea during Permian times, based on palaeomagnetic data, result in major overlaps between Gondwana and Laurasia when choosing a Wegener type configuration (Pangaea A1 or A2). In order to reduce these overlaps, several alternative reconstructions for Pangaea have been proposed whereby Gondwana is shifted towards the east with respect to Laurasia (Pangaea B and Pangaea C). Although these configurations are perfectly compatible with the palaeomagnetic data set, they are not accepted by the majority of the earth science community. Gondwana's palaeomagnetic data set is of rather low quality and often hampered by large uncertainties in rock age. The overlap of Laurasia and Gondwana, therefore, may be an artefact. Here we present new palaeomagnetic data for the radiometrically dated (280 ± 2 Ma, single mineral, K–Ar) Jebel Nehoud ring complex, Northern Kordofan, Sudan. The majority of the samples studied show rather simple directional behavior. After removal of component A, pointing to the north with shallow inclinations and bearing great similarity to the direction of the present day Earth's magnetic field in the region, component B was identified at demagnetization temperatures above 380 °C. Maximum unblocking temperatures for component B reach 580 °C and only occasionally 650 °C or higher, suggesting magnetite and occasionally haematite as the principal carriers of component B. Component B is of single polarity and points towards the southeast with intermediate inclinations. The resulting overall mean south pole position based on eight sites is 40.8°S and 71.3°E (k: 78.1; α_{95}: 6.0) and 46.5°S; 68.0°E when transferred into northwest African coordinates. Combining this new pole position with the highest quality data from Australia, South America and Africa results in a mean pole position at 37.3°S; 63.8°E ($N = 11$, $k = 84.5$, $\alpha_{95} = 5.0$) in northwest African coordinates and removes some of the ambiguities in the Gondwana data set for the Early Permian. However, the palaeogeographic reconstruction of Gondwana on the basis of the new data, results in a major overlap with Laurasia when a Pangaea A2 configuration is chosen, whilst it is fully compatible with a Pangaea B type configuration.
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Keywords: Palaeogeographic reconstructions; Configuration; Magnetic field

1. Introduction

The configuration of Pangaea during Permian and Triassic times is still a matter of debate (see McElhinny and McFadden, 2000 and references therein, Van der Voo, 1993), and at least four different palaeogeographic reconstructions have been proposed over the years, ranging from Pangaea A1 and A2 to Pangaea B and Pangaea C (Fig. 1). Reconstructions of type A1 (Bullard et al., 1965) and A2 (Van der Voo and French, 1974), are only minor modifications of Wegener's (1915) original reconstruction, where South America is facing North America and Africa is facing Europe and the southeastern coast of North America. Reconstructions of type B (Morel and Morel, 1981) and C (Smith and Livermore, 1991), however, are significantly different. Here, the northern margin of Gondwana is in much higher latitudes and South America is facing Europe, requiring a relative eastward translation of Gondwana of several thousand kilometers when compared to models of A1 and A2 type. Although, at first sight, this seems to be a rather unlikely scenario and is presently not backed up by geological data, it should be pointed out that Pangaeas of type B and C are fully compatible with palaeomagnetic data. Reconstructions of type B and C, however, are not widely accepted by the earth science community, the main reasons being that there is an almost unanimous agreement (McElhinny and McFadden, 2000; Van der Voo, 1993) that the opening

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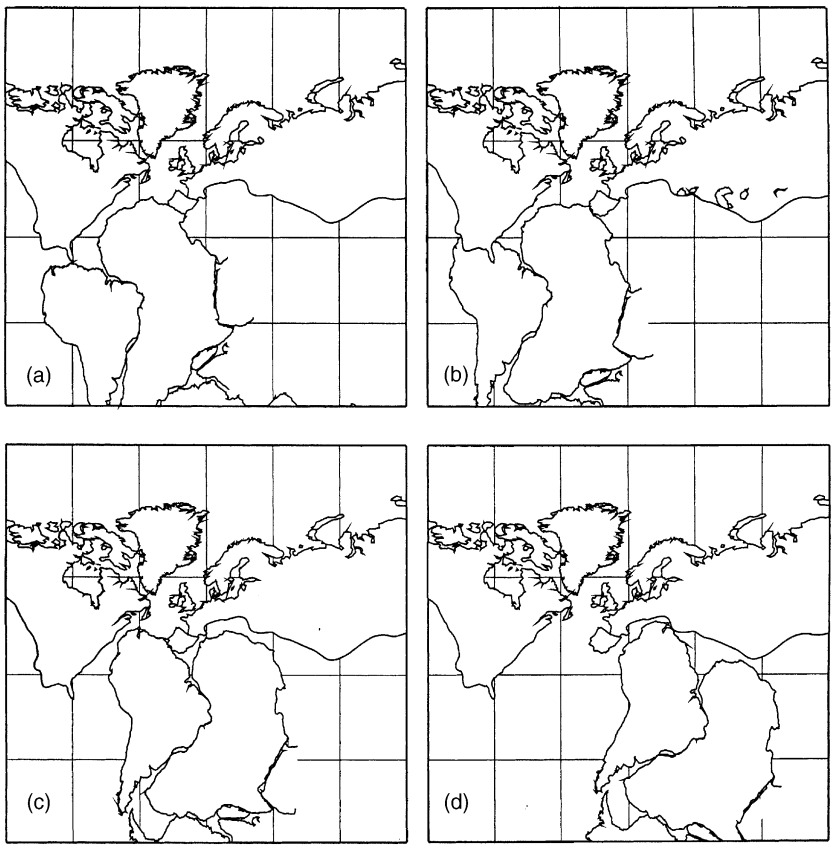


Fig. 1. Palaeogeographic reconstructions of Gondwana in Early Permian times with North America fixed in present day coordinates. (a) Pangaea A1 (Bullard et al., 1965), (b) Pangaea A2 (Van der Voo et al., 1976), (c) Pangaea B (Morel and Morel, 1981), (d) Pangaea C (Smith et al., 1981).

of the Atlantic ocean in Early Jurassic times (Klitgord and Schouten, 1986) started from a Pangaea A type configuration. If a Permian Pangaea B or C is adopted, this requires dextral strike slip of Gondwana in the order of several thousand kilometers with respect to Laurasia prior to Early Jurassic times (McElhinny and McFadden, 2000). Horizontal motions of this predicted order of magnitude require major shear zones between Laurasia and Gondwana, the evidence for which are very sparse at best. However, whilst the Late Palaeozoic/Early Mesozoic palaeomagnetic data set for Laurasia is now fairly well established, the data set for Gondwana is still rather poor. The statistical errors associated with the resulting mean poles for Carboniferous to Permian times are unacceptably high and the uncertainties surrounding the age of individual palaeopole positions for Gondwana are rather large (Van der Voo, 1993). Pangaea B and C, therefore, could be artefacts due to the generally poor quality of the palaeomagnetic data set for Gondwana. It has also been suggested that large and persistent non-dipole components in the southern hemisphere of the Permian world may have led to erroneous inclination values (Briden et al., 1970), a concept recently revived by Van der Voo and Torsvik (2001). However, recently published high quality results for Permian and Triassic rocks from the southern Alps (Muttoni et al., 1996), the Arabian peninsula (Torcq et al., 1997), Tanzania (Nyblade et al., 1993) and Zimbabwe and South Africa (Opdyke et al., 2001) yield palaeopole positions which lend further support to Pangaea B and C concepts. Addressing these questions we have studied a well dated ring complex (280 Ma, K/Ar single minerals (Müller-Sohnius and Horn, 1994)) in Northern Kordofan, Sudan.

2. Geologic setting of the Jebel Nehoud ring complex and palaeomagnetic sampling

The size and shape of igneous ring complexes vary substantially and range from circular to elliptical with diameters between less than 1 to 30 or more kilometers. Similar to their variation in shape and size, the petrological composition ranges from carbonatites, gabbros, syenites to granites and their effusive equivalents (Kinnaird and Bowden, 1987 and references therein). So far the occurrence of more than 100 ring complexes has been reported from northeast Africa, where they intrude the metamorphic pre-Cambrian (Pan-African) basement as well as the Palaeozoic and Mesozoic sedimentary

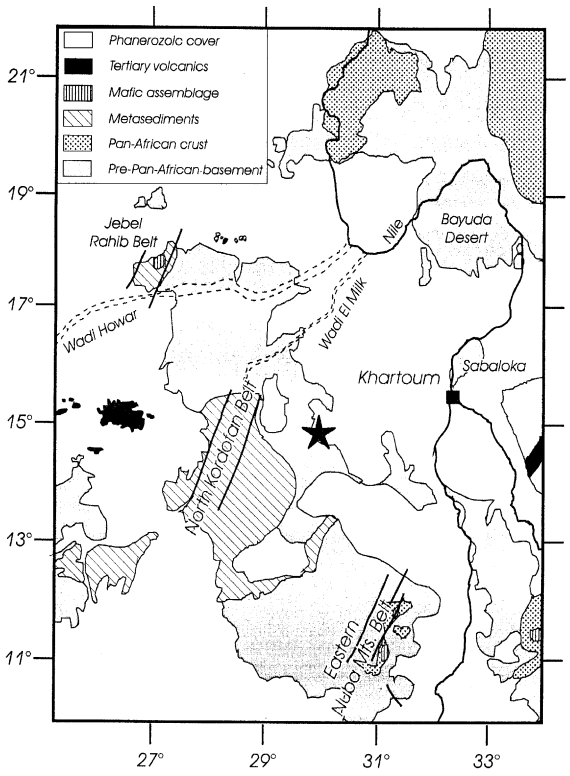


Fig. 2. Simplified geological map of Central Sudan after Abdel Rahman et al. (1990). Jebel Nehoud is marked by a closed star.

cover (Vail, 1976). However, as yet no systematic pattern in terms of the temporal and spatial distribution of ring complexes has been identified (Vail, 1989).

Since Delany (1955) who reported for the first time the occurrence of alkali syenites and rhyolites from a number of isolated complexes in Northern Kordofan (Fig. 2), some 30 igneous ring complexes have been identified and mapped in Sudan in detail by several workers (Vail, 1985 and references therein). However, the temporal relationships between individual complexes are still unclear and absolute age dates remain sparse. Absolute ages for only three complexes are available for the region and, based on Rb–Sr mineral isochron data, yield ages between 246 and 550 Ma (Harris et al., 1983; Heimbach and Curtis, 1980).

Jebel Nehoud, the subject of this study, is situated in Central Northern Kordofan (Fig. 2) and comprises three syenite bodies, which crop out in a N–S trending region of 3.5 km extent. Samples from Jebel Nehoud yield consistent single mineral K/Ar radiometric ages based on both biotite and hornblende of 280 ± 2 Ma (Müller-Sohnius and Horn, 1994), i.e. earliest Permian using the timescale of Harland et al. (1990).

A total of 129 standard palaeomagnetic drill cores (12 sites) were collected using a portable petrol powered drill, covering each of the three sub-complexes labeled Jebel Nehoud North (JNN), Jebel Nehoud South (JNS) and Jebel Nehoud Horseshoe (JNH). A standard magnetic compass was used for the orientation of the samples. No corrections were made for the declination of the local Earth's magnetic field in the region (0.5° during the time of fieldwork).

3. Palaeomagnetic methods

Samples were cut into standard 1-in. specimens. Direction and intensity of the natural remanent magnetization (NRM) was measured using a MOLSPIN spinner magnetometer. Thermal and alternating field (AF) demagnetization experiments were performed in a magnetically shielded thermal demagnetizer (Schoenstedt, TSD-1) and a single axis AF-demagnetizer (ambient field <5 nT). Demagnetization results for each specimen were plotted on orthogonal vector diagrams (Zijderveld, 1967) and on equal-area stereographic projections. Mean directions have been calculated after Fisher (1953).

4. Palaeomagnetic results

Approximately 50% of the samples studied display rather simple and uniform directional behavior (Fig. 3). The remaining samples did not reveal any stable direction of the NRM and will not be discussed further.

Detailed pilot studies, using both AF- and thermal-demagnetization techniques demonstrated that thermal demagnetization was more efficient in resolving the whole directional spectrum of magnetization. AF-demagnetization experiments with maximum peak fields of 80 mT were successful in identifying a southwesterly and downward pointing direction of magnetization but failed to resolve the direction of a second component carried by a high coercivity phase (Fig. 3a).

Thermal demagnetization at temperatures of up to 300 °C removes a northerly and rather steep component of magnetization (component A). Above 300 °C component B which points southwesterly and downward was identified (Fig. 3b–d). This component is directed towards, or reaches the origin of the orthogonal projection. Maximum unblocking temperatures (T_B) vary, but only occasionally exceed 580 °C, thus indicating the presence of magnetite plus minor amounts of a high blocking temperature mineral, probably haematite. Component B is identified in both magnetite and haematite dominated samples. The resulting site mean directions for component B are shown in Fig. 4 and listed in Table 1. The overall mean direction for component B based on eight sites is 147° declination and 39.6° inclination (k: 78.1; α_{95}: 6.0) yielding a palaeopole position at 40.8°S; 71.3°E, and 46.6°S; 68.0°E when transferred into northwest African coordinates applying the rotation parameters given by Lottes and Rowley (1990). We note that this pole position does not show

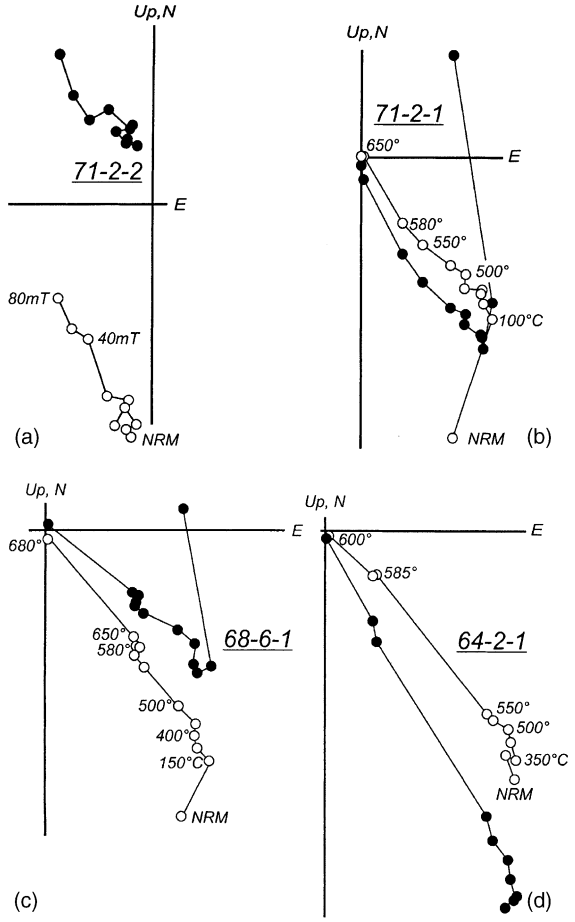


Fig. 3. Orthogonal projection (Zijderveld, 1967) of AF- (a) and thermal (b–d)-demagnetization experiments of representative samples. Note that although AF-demagnetization does not reveal the full directional spectrum, nevertheless, the vector removed between 20 and 80 mT is comparable to the stable endpoint directions in b, c and d. Open (closed) circles are projections on the vertical (horizontal) plane.

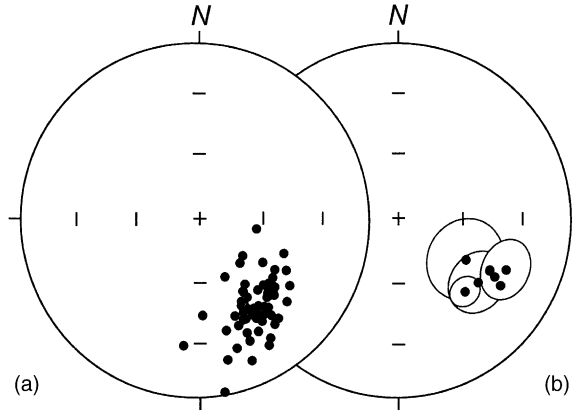


Fig. 4. Stereographic projection (lower hemisphere) of the characteristic sample directions (a) and the resulting site mean directions (b).

any similarities to other poles for Gondwana of younger age, and is interpreted to reflect a primary magnetization acquired during cooling of the ring complex.

Table 1
Site mean directions for component B from Jebel Nehoud

Site	N/n	Dec.	Inc.	K	α_{95}
60	11/6	141.6	41.3	24.0	13.9
61	12/0	–	–	–	–
62	10/0	–	–	–	–
63	10/7	149.2	35.9	45.3	9.1
64	11/11	147.1	30.8	58.0	6.1
65	10/0	–	–	–	–
66	11/8	150.9	39.8	66.0	6.9
67	11/0	–	–	–	–
68	11/7	149.0	52.8	11.7	18.4
69	10/8	132.7	43.1	63.4	7.0
70	10/7	150.6	39.7	53.9	8.3
71	12/8	154.9	32.2	20.1	12.7
Site mean direction	12/8	147.2	39.6	78.1	6.0
Resulting pole position		40.8°S	71.3°E		

N/n: number of samples measured/number of samples used for statistics. Dec., Inc.: magnetic declination and inclination, respectively. Also given are the statistical parameters α_{95} and k according to Fisher (1953). The resulting mean pole based on 8 out of 12 sites is given as a south pole.

5. Rock magnetic properties

Based on the results of thermal and AF-demagnetization, a low coercive, intermediate (<580 °C) unblocking temperature magnetic mineral (magnetite) has been identified as the principle carrier of the high temperature component. This is confirmed by most of the isothermal remanent magnetization (IRM) acquisition curves (Fig. 5a), thermal demagnetization of saturation remanence (Fig. 5b), Curie point determinations (Fig. 5c) and X-ray diffraction studies (Fig. 5d) which yield lattice constants between $a_0 = 0.8381$ and 0.8398 nm and are diagnostic for magnetite ($a_0 = 0.839$ nm). The significant inflection of intensity decay curves during thermal demagnetization at 350 °C (Fig. 5b) in conjunction with the shape of the J_S–T curves (Fig. 5c) is indicative for the mineralogical break down of maghaemite as an accessory magnetic phase. Samples from sites 68 to 71 display maximum unblocking temperatures well above 580 °C (Fig. 5b) in combination with rather high coercivities (Fig. 5a) indicating the presence of haematite. The directional characteristics of these haematite bearing samples upon demagnetization above 580 °C (Fig. 3b and c), does not significantly differ from samples where magnetite has been identified as the sole carrier of the magnetization (Fig. 3d). This strongly suggests that both magnetite and haematite acquired the magnetization simultaneously. Reflected light microscopy, indicating the high temperature oxidation state (deuteric formation) of haematite lends further support to this interpretation.

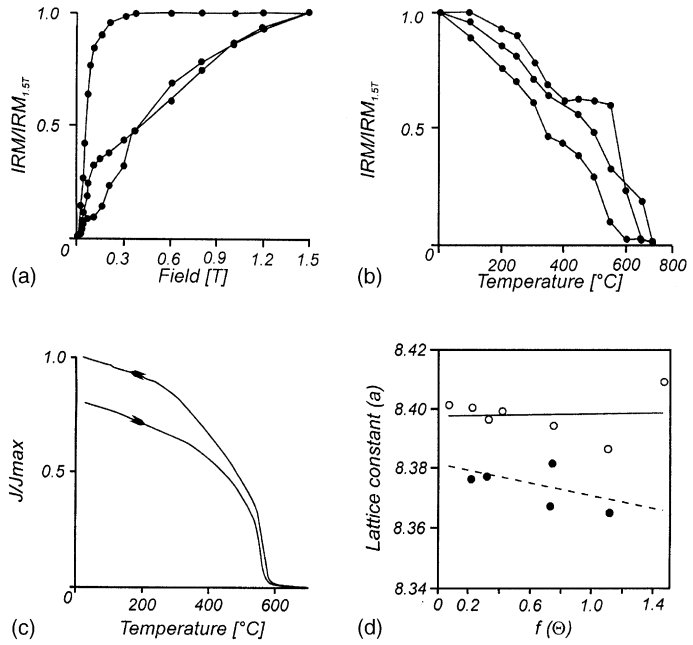


Fig. 5. (a) Normalized IRM acquisition curves in fields up to 1.5 T for three representative samples showing varying amounts of magnetite and haematite. (b) Thermal demagnetization of IRM acquired in field of 1.5 T. (c) Example for thermomagnetic behavior of a magnetite dominated sample. (d) Determination of the lattice constant for samples from two different intrusive bodies. The lattice constants determined are 8.398 and 8.381.

6. Comparison with previously published Early Permian pole positions for Gondwana

Despite the rather high number of palaeomagnetic results for the Permo-Carboniferous of Gondwana, there is still some uncertainty about shape and the fine structure of the apparent polar wander path (APWP) for the southern continents for this time window. This may reflect a combination of several factors: poor quality of some of the published data, poor age control, inconsistencies between the Permo-Carboniferous data sets for East Gondwana (Madagascar, India, Antarctica, and Australia) and West Gondwana (South America, Africa), and problems with the palaeogeographic fits of the Gondwana continents (McElhinny and McFadden, 2000; Van der Voo, 1993).

A investigation of the Global Palaeomagnetic Database (GPMDB, McElhinny, 1998) version 3.5 released June 2000 yields a total of 45 pole positions for Africa, South America and Australia for the Early Permian time interval of 256–290 Ma (Harland et al., 1990) which have been interpreted by the original authors to be of primary origin. In order to judge the quality of the data set, the reliability scheme of Van der Voo (1993) has been used. As we are looking at a very narrow time window (e.g. latest Carboniferous to earliest Permian) the first Van der Voo criterion (whereby the age is known to within half a period) is considered by the current authors as being critical. In addition, at least two of the criteria which require sufficient number of samples and adequate statistics, vector analysis of demagnetization data, or positive field tests have to be fulfilled before a pole is deemed reliable. For Africa, this reduces the data set to six results. Recently published results for the Dwyka varves of Permo-Carboniferous age from southern Africa (Opdyke et al., 2001) have been included. This new pole, supersedes the palaeopole position for the K1 Dwyka varves published earlier (McElhinny and Opdyke, 1968). Importantly, the ambiguity caused by the fact that the original K1-pole is displaced by some 30° westward with respect to other African data of similar age has been removed by these new results.

Three out of four palaeopole positions published for the Early Permian of Australia pass the quality filter (Table 2) and are of very high quality as far as both age determinations and the palaeomagnetic data are concerned. A positive contact test for the Mount Leyshon intrusive complex (Clark, 1996) indicates the primary character of the identified direction.

The same selection criteria have been applied to the South American data set. Here, only two results (Rapalini and Vilas, 1991, Tomezzoli and Vilas, 1999) out of seven fulfill the selection criteria as defined above. The Early Permian age of the palaeopole for the Tunas Formation (Tomezzoli and Vilas, 1999), is supported by a positive fold test after McFadden (1990) at 30% unfolding. Since it has been demonstrated that the Tunas

Table 2
Lower Permian palaeopole positions (south poles) for Australia, Africa and South America rotated into northwest African coordinates (Lottes and Rowley, 1990)

Location	Age (My)	Palaeopole position		k	α_{95}	Tests	Refno	Reference
		Lat. (°S)	Long. (°E)					
Africa								
Chougrane red beds	256–290	32.2	64.1	20.0	4.7	F+	723	Daly and Pozzi (1976)
Upper El Adeb Larache Formation	270–303	38.5	57.5	837.0	2.8		2540	Henry et al. (1992)
Lower Tiguentourine Formation	280–300	33.8	61.4	170.0	4.1		2728	Derder et al. (1994)
Abadla Formation Lower Unit	260–290	29.1	57.8	13.8	3.6		3275	Merabet et al. (1998)
Dwyka Formation	260–303	31.6	63.6		12.0	F+		Opdyke et al. (2001)
Jebel Nehoud	280 ± 2[a]	46.4	68.0	78.1	6.0			this study
Australia								
Mount Leyshon	286 ± 6[a]	35.4	71.6	53.5	3.4	C+	3262	Clark (1996)
Featherbed Volcanics	280 ± 12[b]	31.9	69.3	170.6	5.1		3266	Klootwijk and Giddings (1993)
Tuckers Igneous Complex	283 ± 9[a] 287 ± 4[c]	41.0	69.2	40.9	3.6		3262	Clark (1996)
South America								
Tambillos Formation	256–280	55.5	56.7	50.6	5.2	Fo	2475	Rapalini and Vilas (1991)
Tunas Formation	260–290	33.9	65.0	47.3	4.8	F+	3293	Tomezzoli and Vilas (1999)
Mean pole position	Early Permian	37.3	63.8	84.5	5.0			

Lat., Latitude; Long., Longitude; Refno refers to the reference number in the Global Palaeomagnetic Database (McElhinny, 1998). Ages are stratigraphic ages unless otherwise stated and based on the timescale given by McElhinny (1998). Also given are the statistical parameters α_{95} and k according to Fisher (1953). F: foldtest; C: contact test. Positive field tests are denoted by a "+" and inconclusive results by "o".

[a] K–Ar age.
[b] Rb–Sr age.
[c] U–Pb age.

Formation underwent syndepositional deformation, this result very likely reflects the direction of the Earth's magnetic field during Early Permian times. We note that the statistically well defined palaeopole for the La Tabla Formation, Chile (Jesinkey et al., 1987), supported by a positive conglomerate test does not pass our first requirement (precise age control within half a period) and, therefore, has not been included in our selection.

Combining the data for Africa, South America and Australia listed in Table 2 with the result obtained in this study yields a mean palaeopole position of 37.3°S and 63.8°E when transferred into NW African coordinates (Lottes and Rowley, 1990). This result is not significantly different from the Gondwana mean pole compiled recently by McElhinny and McFadden (2000) and the West Gondwana mean pole published by Muttoni et al. (1996), Van der Voo (1993) and Nyblade et al. (1993) for the Early Permian based on sedimentary sequences for which relative age constraints are often rather poor.

This is the first pole position for this time interval derived from from radiometrically dated rocks from western Gondwana, thus removing the question whether poor age control of the African palaeopoles is responsible for the overlap between Gondwana and Laurasia in a classical Pangaea A1 configuration.

7. Conclusion

Although no field tests could be carried out in order to constrain the timing of the acquisition of the ChRM, the rock magnetic properties strongly suggest a primary origin. No evidence for significant alteration other than deuteric formation of haematite and/or maghemite is evident. Therefore, the time of acquisition of the ChRM in the Jebel Nehoud complex is considered to be linked with the closure of the K/Ar isotope system in biotite and hornblende at 280 ± 2 Ma (Müller-Sohnius and Horn, 1994). If this line of argument is correct, we now have the first palaeopole position for the Permo-Carboniferous of Africa which is supported by an absolute age assignment. However, it is worthwhile to point out that the new result from Jebel Nehoud plots 14° away from the cluster of poles from NW and central-East Africa (Table 2), but in turn is in rather good agreement with the equally well dated results for igneous rocks from Australia. It could be argued that the discrepancy

between the result from Jebel Nehoud, and the other pole positions from West Africa, reflects substantial local tectonic rotations. Indeed the study area is situated close to the postulated eastward extension of the Central African Fault Zone (CAFZ; Schandelmeier and Pudlo, 1990; and Schandelmeier and Richter, 1991) which separates the West Sahara from the East African block. Jebel Nehoud is situated in the Umm Badr Shear Zone (UBSZ) which developed during late Pan-African crustal shortening and which was reactivated in the Carboniferous (Schandelmeier and Pudlo, 1990; Schandelmeier and Richter, 1991). Some arguments favour another phase of reactivation during Carboniferous to Triassic times but there is, as yet, no geological evidence for any post-Permian translations and/or rotations in the region. Alternatively, and the argument preferred here, given the close similarity of the new pole with coeval data from Australia, is that the differences between the Jebel Nehoud pole and the other West African poles reflects uncertainties in stratigraphic correlation. All West African results have been derived from sedimentary sequences dated biostratigraphically. More reliable palaeomagnetic determinations in combination with reliable, preferably absolute, age determinations are required before a final solution of this problem can be attempted.

Reconstructions of Pangaea based on the new mean pole for Gondwana and the well established pole position for the Early Permian for Laurasia at 46°S, 301°E (Van der Voo, 1993) are fully compatible with Pangaea B type reconstructions (Fig. 6a) and no significant overlaps between Laurasia and Gondwana are observed. Alternatively, we tested our results against a Pangaea A2 (Van der Voo et al., 1976). The Pangaea A2 reconstruction for pre-Jurassic times (Van der Voo and French, 1974) were made by comparing the distribution of palaeomagnetic poles from Laurasia to those from Gondwana after the latter had been rotated into the North American coordinate system (Bullard et al., 1965). However, the Bullard rotation parameters are not sufficient to match the palaeomagnetic data from Laurentia and Gondwana. To bring the sets of palaeopoles from Gondwana into agreement with the one from Laurentia, they applied a 20° clockwise rotation about an Euler pole 27°N and 300°E. Consequently Van der Voo and French (1974) argued that Gondwana has to be rotated by 20° cw with respect to its position in the classical Bullard fit (Pangaea A1). We note however that the position of Gondwana in their reconstruction for the Upper Carboniferous, the Lower and the Upper Permian are not based on actual data for this continent. The pole positions for Gondwana as listed in their Table 2 were determined by transferring the Laurentian pole positions into African coordinates followed by a second rotation as outlined above. Following the procedure outlined by Van der Voo and French (1974), rotating the new Gondwana mean pole into the North American coordinate system (Bullard et al., 1965) followed by clockwise rotation by 20° about a pole at 27°N and 300°E places the northern margin of Gondwana approximately 15° further north than in the original A2 reconstruction and results therefore in unacceptable overlaps between the east coast of North America and the West Africa as well as between North Africa and Europe (Fig. 6b).

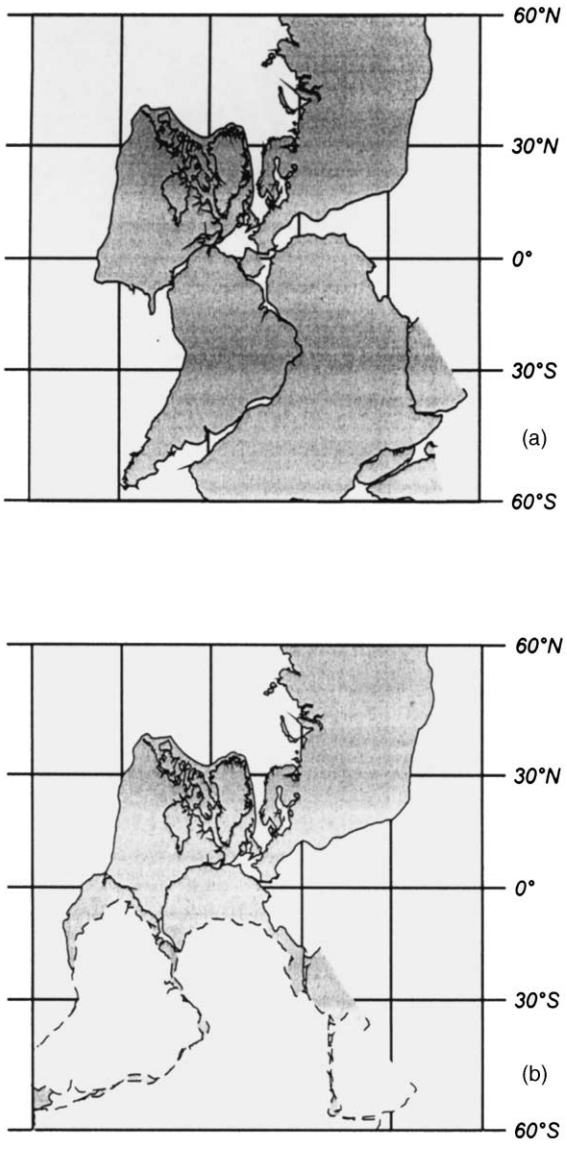


Fig. 6. Reconstruction of the Atlantic bordering continents in Early Permian times based on the mean pole for Gondwana (Table 1) and the reference pole for Laurasia (46°N, 124°E), taken from Van der Voo (1993, Table 5.7) in North American coordinates. (a) Pangaea B type reconstruction, (b) Pangaea A2 type reconstruction, resulting from rotation of the new mean pole for Gondwana (Table 1) into the North American reference system (Bullard et al., 1965) followed by a clockwise rotation of 20° around a pole at 27°N and 300°E (Van der Voo and French, 1974). Gondwana in the original A2 configuration (Van der Voo and French, 1974) is shown in white and marked by the stippled line.

Accepting that there is sufficient evidence for the primary character of the magnetization determined for the Jebel Nehoud, problems in age assignments for the African, South American and Australian pole positions used here to calculate an Early Permian mean pole can be ruled out. The remaining argument, i.e. non-dipole fields, cannot be addressed directly by our results as the magnetizations are of single polarity. So far, no dual polarity magnetizations have been reported for the Kiaman Geomagnetic Superchron of inverse polarity, which lasted from 298–262 Ma (Opdyke and Channell, 1996) and thus no reversal tests have been possible to constrain any possible non-dipole effects. Magnetizations with mixed polarities, however, have been reported for lowermost Triassic sediments (Embleton and McDonnell, 1980) and ignimbrites (Lackie, 1988) of uppermost Permian age from Australia. Both results pass the reversal test with classification "c" (angular difference between normal and reversed polarities less than 10°, McFadden and McElhinny (1990)). This can be taken as evidence that at least during uppermost Permian times the Earth's magnetic field was clearly dominated by dipolar components and that non-dipole field contributions only played a minor role.

Although, far from solving the problems of Pangaean palaeogeography in the Permian, the results from the Jebel Nehoud, in combination with data for presumably coeval rocks from elsewhere in Africa and Australia, reduce some of the previously discussed ambiguities in the Permian palaeomagnetic data set. The answer to the question—"which configuration for Pangaea to choose in Early Permian times", therefore must be based on geological arguments.

Acknowledgements

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